ASSIGNMENT – 3

MCA | Machine Learning

Complete Questions with Answers

Course	Machine Learning
Programme	Master of Computer Applications (MCA)
Last Date	9 October 2026
Total Questions	10 (5 marks each)

Instructions: All 10 questions are included, followed by clear, question-wise answers with formulas and examples.

Question 1 (5 marks)

Question: Define Supervised Learning. Explain the key difference between supervised learning and unsupervised learning with one example of each.

Answer: Supervised learning is a machine-learning approach in which a model learns from labelled examples. Each training example contains input features and a known target/output. The model learns a relationship between inputs and outputs so that it can predict outputs for new data.

Difference: In supervised learning, the data includes the correct labels or target values; in unsupervised learning, data has no target labels and the model looks for patterns or groups on its own.

Example of supervised learning: A spam filter is trained using emails labelled “spam” or “not spam” and predicts the category of a new email.
Example of unsupervised learning: A shop groups customers into segments based on purchase behaviour using clustering, without pre-assigned customer-group labels.

Supervised learning is commonly used for classification and regression, while unsupervised learning is commonly used for clustering, association, and dimensionality reduction.

Question 2 (5 marks)

Question: What are the two main types of supervised learning problems? Clearly distinguish between them and give one real-world example for each type.

Answer: The two main types are **classification** and **regression**.

1. Classification: Predicts a discrete class or category. Examples include spam/not spam, approved/rejected, or healthy/at-risk. A suitable algorithm is Logistic Regression, Decision Tree, or Support Vector Machine.
Real-world example: A bank predicts whether a loan application should be classified as “high risk” or “low risk.”

2. Regression: Predicts a continuous numerical value. Examples include price, temperature, sales, or travel time. A suitable algorithm is Linear Regression, Random Forest Regressor, or another regression model.
Real-world example: A real-estate model predicts a house price in rupees.

Main difference: Classification outputs a label/category; regression outputs a numerical quantity.

Question 3 (5 marks)

Question: Explain the concept of Linear Regression. Write the equation of a simple linear regression model and define the terms (dependent variable, independent variable, intercept, and slope). What is the role of the residual/error term?

Answer: Linear Regression is a supervised learning technique that models the relationship between a numerical dependent variable and one or more independent variables by fitting a line (or, in multiple dimensions, a linear function) to observed data.

Simple linear regression equation:
y = β■ + β■x + ε

y (dependent variable): The value we want to predict, such as house price. **x (independent variable):** The input feature used to predict y, such as house area. β■ **(intercept):** The predicted value of y when x = 0. β■ **(slope):** The expected change in y for a one-unit increase in x. ε **(error term):** Represents variation not

explained by the model. A fitted model produces a prediction ■ = β■ + β■x. The **residual** for an observed data point is e = y – ■, the difference between the actual and predicted value. Residuals help assess model fit and identify patterns or unusual observations.

Question 4 (5 marks)

Question: How does Logistic Regression differ from Linear Regression? Write the logistic (sigmoid) function and explain why it is suitable for binary classification. What is the decision boundary in logistic regression?

Answer: Linear Regression predicts a continuous numerical value, whereas Logistic Regression estimates the probability that an example belongs to a class and is commonly used for binary classification.

The logistic (sigmoid) function is:
$\sigma(z) = 1 / (1 + e^{-z})$
where z = β■ + β■x■ + … + β■x■. The sigmoid converts any real-valued z into a value between 0 and 1, which can be interpreted as an estimated probability for the positive class.

For example, if the estimated probability of a transaction being fraudulent is 0.82, a threshold of 0.5 would classify it as fraud. The threshold can be changed depending on the application's needs.

Decision boundary: The boundary separates the feature space into predicted classes. With a 0.5 probability threshold, it occurs where σ(z) = 0.5, equivalent to z = 0. The boundary is linear in the original features unless features or transformations make it nonlinear.

Question 5 (5 marks)

Question: Describe how a Decision Tree works for classification. Explain the concepts of Information Gain and Gini Impurity. Why can decision trees overfit the training data, and how can this problem be reduced?

Answer: A Decision Tree classifies data by repeatedly splitting it according to feature-based rules. The top node is the root, internal nodes represent tests, branches represent outcomes, and leaf nodes provide the predicted class. For example, a tree may classify a loan application using income, credit history, and existing debt.

Information Gain: Measures the reduction in entropy after splitting a dataset on a feature. A feature with higher information gain creates a greater improvement in class separation.
Gini Impurity: Measures how mixed the classes are in a node. For class proportions p■, Gini = 1 – Σp■². A value of 0 means the node contains only one class.

A tree may overfit when it grows too deep and learns noise or rare details from the training data. It may then perform poorly on new data. Overfitting can be reduced by limiting maximum depth, requiring a minimum number of samples per split or leaf, pruning branches, selecting useful features, and validating the model with cross-validation.

Question 6 (5 marks)

Question: What are Ensemble Methods? Differentiate between Bagging and Boosting with suitable examples of algorithms that use each technique. Why do ensemble methods generally perform better than a single model?

Answer: Ensemble methods combine predictions from multiple models to produce a final prediction that is often more accurate or stable than that of a single model.

Bagging (Bootstrap Aggregating): Trains models independently on different bootstrap samples of the training data and combines their outputs. For classification, voting may be used; for regression, predictions are often averaged. It mainly reduces variance. **Example:** Random Forest, which combines many decision trees.

Boosting: Trains models sequentially, with each new model focusing more on errors or difficult examples from earlier models. The final prediction combines the learners. **Examples:** AdaBoost, Gradient Boosting, and XGBoost.

Key difference: Bagging models can be trained independently and then aggregated; boosting builds learners in sequence to improve previous performance.

Ensembles can perform better because different models capture different patterns, averaging can reduce instability, and sequential correction can reduce systematic errors. Improvement is not guaranteed; it depends on data, model choice, and tuning.

Question 7 (5 marks)

Question: Explain the working principle of Support Vector Machines (SVM) for binary classification. What are support vectors and the margin? Briefly describe the role of the kernel trick in SVM.

Answer: A Support Vector Machine (SVM) finds a separating hyperplane that divides examples from two classes. Among possible separating hyperplanes, a standard maximum-margin SVM selects one that maximises the distance to the closest training examples from each class.

Support vectors: The training examples closest to the decision boundary. They determine the position and orientation of the maximum-margin boundary.
Margin: The distance between the decision boundary and the nearest examples from either class. A larger margin can support better generalisation, although noisy data may require a soft-margin approach.

Kernel trick: When classes cannot be separated well by a straight line in the original feature space, a kernel function computes relationships corresponding to a higher-dimensional feature space without explicitly calculating every transformed feature. Common kernels include linear, polynomial, and radial basis function (RBF). The chosen kernel and its parameters affect model performance.

Question 8 (5 marks)

Question: List and briefly explain any four-evaluation metrics used for classification problems (e.g., Accuracy, Precision, Recall, F1-Score, ROC-AUC). When is Accuracy not a good metric, and which metric would you prefer instead?

Answer: Common classification evaluation metrics include:

1. Accuracy: Proportion of all predictions that are correct. Formula: (TP + TN) / (TP + TN + FP + FN).
2. Precision: Of all examples predicted positive, the proportion that are truly positive. Formula: TP / (TP + FP).
3. Recall (Sensitivity): Of all actual positive examples, the proportion correctly identified. Formula: TP / (TP + FN).
4. F1-Score: Harmonic mean of precision and recall. Formula: 2 × (Precision × Recall) / (Precision + Recall).
5. ROC-AUC: Measures how well a classifier ranks positive examples above negative examples across classification thresholds; larger values generally indicate better ranking ability.

Here TP = true positives, TN = true negatives, FP = false positives, and FN = false negatives.

Accuracy may be misleading on **imbalanced datasets**. For example, if only 1% of transactions are fraudulent, a model that always predicts “not fraud” can achieve 99% accuracy while detecting no fraud. Prefer recall when missing positive cases is costly, precision when false alarms are costly, or F1 when balancing precision and recall is important. PR-AUC can also be useful for rare positive classes.

Question 9 (5 marks)

Question: For regression problems, explain the following evaluation metrics: (a) Mean Squared Error (MSE) (b) Root Mean Squared Error (RMSE) (c) Mean Absolute Error (MAE) (d) R-squared (Coefficient of Determination). Which metric is more sensitive to outliers and why?

Answer: Let y■ be an actual value, ■■ a predicted value, and n the number of observations.

(a) Mean Squared Error (MSE): Average squared prediction error.
MSE = (1/n) Σ(y■ – ■■)²
It penalises large errors strongly because errors are squared.

(b) Root Mean Squared Error (RMSE): Square root of MSE.
RMSE = √MSE
It is expressed in the same units as the target variable and still gives greater weight to large errors.

(c) Mean Absolute Error (MAE): Average absolute prediction error.
MAE = (1/n) Σ|y■ – ■■|
It is easy to interpret in the target’s units and is less affected by a single large error than MSE/RMSE.

(d) R-squared (R^2): Measures how much variation in the target is explained by the model relative to predicting the target mean.
$R^2 = 1 – [\Sigma(y■ – ■■)^2 / \Sigma(y■ – ■)^2]$
Here ■ is the mean of actual values. $R^2 = 1$ indicates a perfect fit; $R^2 = 0$ means performance equivalent to predicting the mean under the usual definition. R^2 can be negative on test data.

Outlier sensitivity: MSE and RMSE are more sensitive to outliers because the errors are squared, making very large errors disproportionately influential. MAE is comparatively more robust.

Question 10 (5 marks)

Question: Discuss any four real-world applications of Supervised Learning. For each application, mention the type of supervised learning problem (classification or regression) and a suitable algorithm that can be used.

Answer:

1. Email spam detection
Problem type: Classification (spam or not spam).
Suitable algorithm: Naive Bayes, Logistic Regression, or Support Vector Machine.

2. House-price prediction
Problem type: Regression (predicting a numerical price).

Suitable algorithm: Linear Regression, Random Forest Regressor, or Gradient Boosting Regressor.

3. Medical risk screening
Problem type: Classification (for example, predicting whether a patient is in a higher-risk category based on labelled data).
Suitable algorithm: Logistic Regression, Decision Tree, or Random Forest. Such predictions support, but do not replace, professional clinical judgement.

4. Sales forecasting
Problem type: Regression (predicting future sales quantity or revenue).
Suitable algorithm: Linear Regression, Random Forest Regressor, or Gradient Boosting Regressor, using appropriate historical and time-related features.

These applications use labelled historical examples to learn a relationship between input features and a known target. Model quality should be evaluated on data not used for training.